

**Evaluating Expressions with Radicals***Question Bank - Additional 4.05*

---

---

*Show your work. Circle your final answer.*

**For questions 1 through 6, evaluate.**

1.  $\sqrt{64}$

2.  $\sqrt[3]{8}$

3.  $\sqrt[3]{-125}$

4.  $\sqrt{\frac{9}{49}}$

5.  $\sqrt{4^2 + 33}$

6.  $\sqrt[3]{5^2 + 100}$

**For questions 7 and 8, evaluate. Each expression has a power and a radical.**

**7.** Evaluate each expression.

**a.**  $\sqrt{144} \cdot 3^{-2}$

**b.**  $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

**c.**  $\sqrt[3]{-27} - 2^{-1}$

**8.** Evaluate each expression.

**a.**  $\sqrt[3]{64} - 1.5(\sqrt{7^2 - 13})$

**b.**  $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

**c.**  $\sqrt{6^2 + 13} \cdot 2^{-3}$

**Working backwards.**

9. Find the value of  $n$  that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a.  $\sqrt{n} = 15$

b.  $\sqrt[3]{n} = -2$

c.  $\sqrt{n-5} = 11$

**Reading someone else's work.**

10. Deshawn writes  $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$  . Evaluate the expression correctly, and explain in one sentence what the radical bar does that his method ignores.

11. Which is the value of  $\sqrt{8^2+36}$  ?

- ☐ A. 14
- ☐ B. 10
- ☐ C. 100
- ☐ D. 50

12. Select ALL of the expressions whose value is 4.

☐ A.  $\sqrt{2^3 + 17}$

☐ B.  $\sqrt[3]{64}$

☐ C.  $\sqrt[3]{-64}$

☐ D.  $\sqrt{16}$

☐ E.  $\sqrt{5^2 - 9}$

☐ F.  $\sqrt[3]{3^3 + 37}$